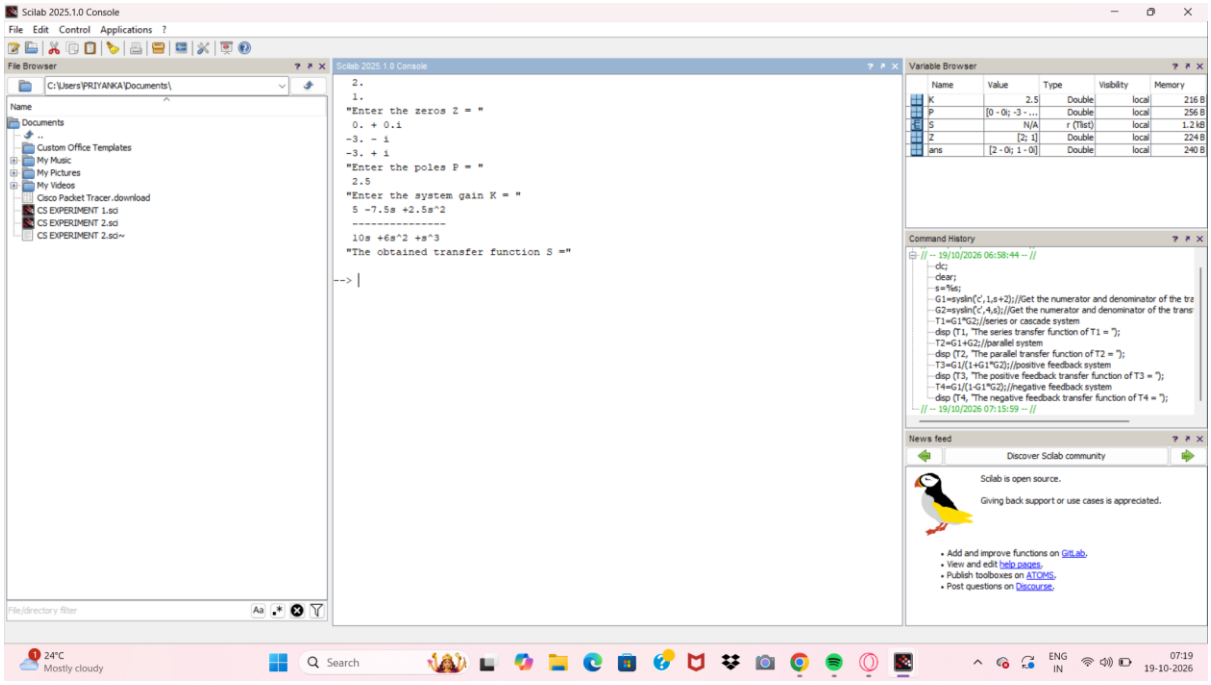
PROGRAM CODE:

(i) Transfer function from the given poles and zeros

```
clc; clear;
Z=[2;1];P=[0;-3-%i;-3+%i];K=2.5;
disp(Z,'Enter the zeros Z = ')
disp(P,'Enter the poles P = ')
disp(K,'Enter the system gain K = ')
S=zp2tf(Z,P,K,"c")
roots(S.num)
disp(S, 'The obtained transfer function S =')
```

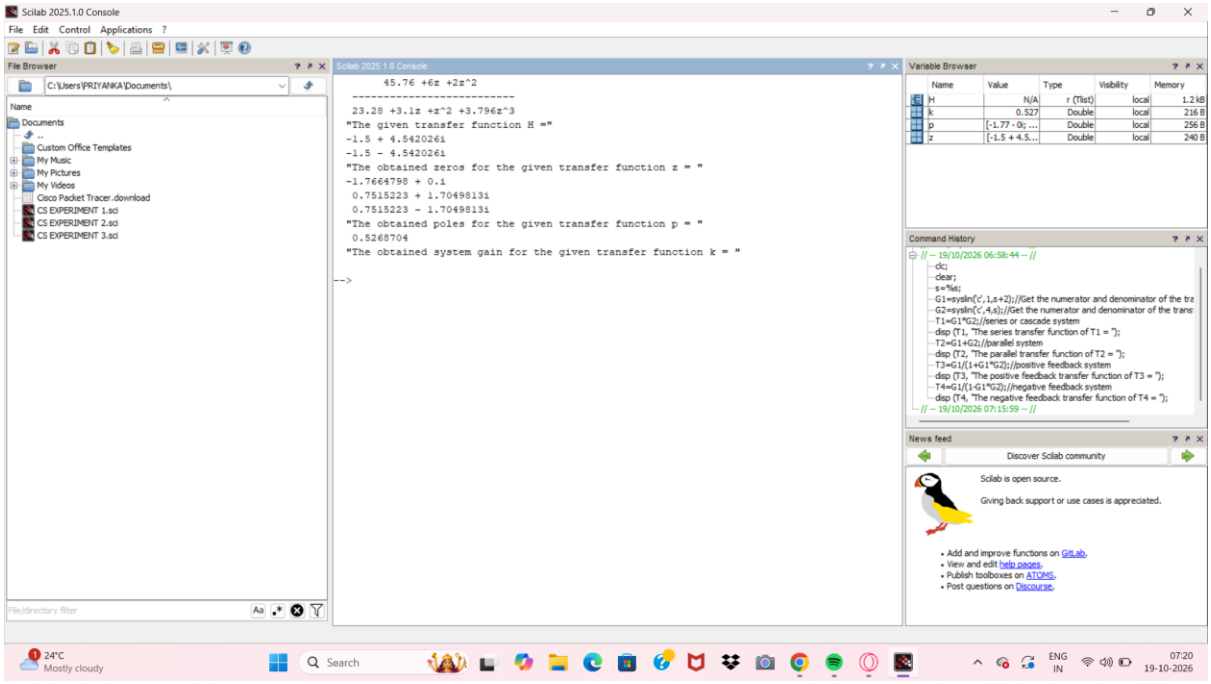
OUTPUT:



(ii) The zeros and poles from the given transfer functions.

```
clc
clear
```

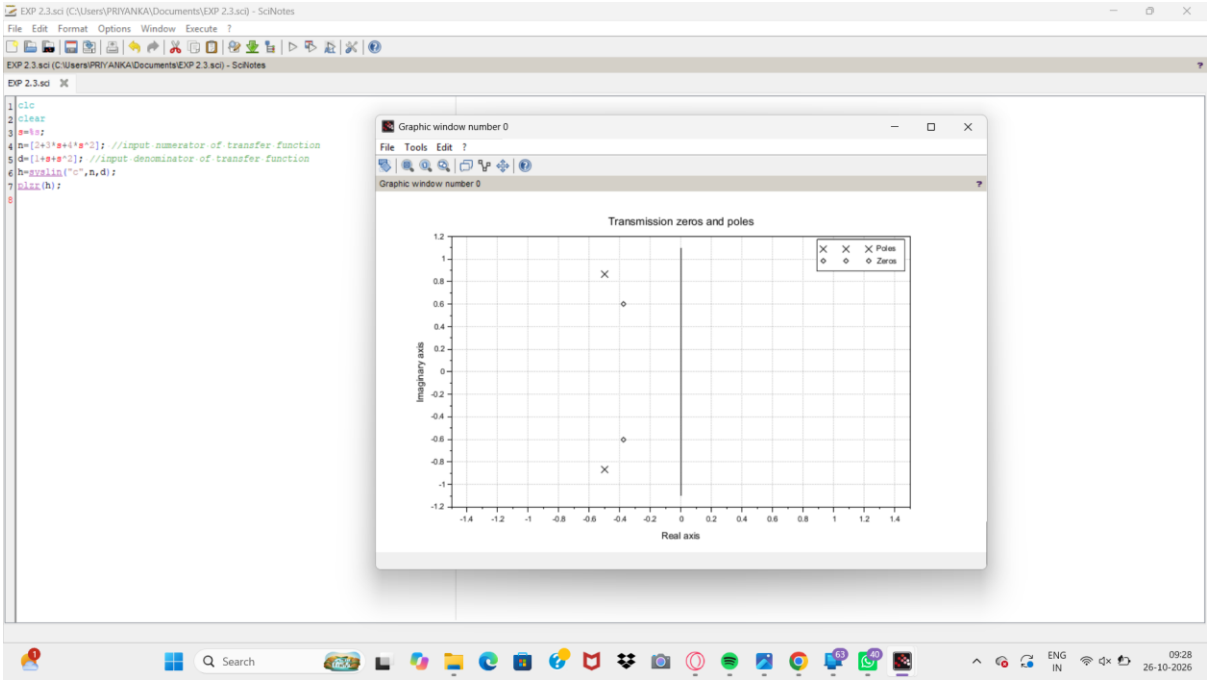
```
H=syslin("d",[45.76+6*%s+2*%s^2],[23.28+3.1*%s+%s^2+3.796*%s^3])
disp(H, 'The given transfer function H =' )
[z,p,k]=tf2zp(H)
disp(z, 'The obtained zeros for the given transfer function z = ');
disp(p, 'The obtained poles for the given transfer function p = ');
disp(k, 'The obtained system gain for the given transfer function k = ');
```



(iii) Plot zeros and poles in the s plane for the given transfer function

```
clc
clear
s=%s;
n=[2+3*s+4*s^2]; //input numerator of transfer function
d=[1+s+s^2]; //input denominator of transfer function
h=syslin("c",n,d);
plzr(h);
```

OUTPUT:


EXP 2.3.sci (C:\Users\PRIYANKA\Documents\EXP 2.3.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 2.3.sci
clc
clear
s=%s;
n=[2+3*s+4*s^2]; //input numerator of transfer function
d=[1+s+s^2]; //input denominator of transfer function
h=syslin("c",n,d);
plzr(h);
Graphic window number 0
File Tools Edit ?
Transmission zeros and poles
Poles
Zeros
Imaginary axis
Real axis
Search
ENG
IN
09:28
26-10-2026